

NETWORK LAYER

Module 3

Problem 3: The IPv4 address and subnet mask are given as 201.123.101.53 and 255.255.255.224, respectively. Find:

- a. the subnet ID for given IP address.
- b. 1st host of the same subnet as given in IP address.
- c. Last host of the same subnet as given in IP address.
- d. Directed broadcast address for the same subnet as given in IP address.
- e. 1st host of the 1st subnet.
- f. DBA for the 1st subnet.
- g. Last host of the last subnet.
- h. DBA for the last subnet.
- i. 3rd host of the 3rd subnet.
- j. DBA for the 3rd subnet.

Solution 3: Do it in class.

Problem 3: The IPv4 address and subnet mask are given as 201.123.101.53 and 255.255.255.224, respectively. Find:

- a. the subnet ID for given IP address. **Ans: 201.123.101.32**
- b. 1st host of the same subnet as given in IP address. **Ans: 201.123.101.33**
- c. Last host of the same subnet as given in IP address. **Ans: 201.123.101.62**
- d. Directed broadcast address for the same subnet as given in IP address. **Ans: 201.123.101.63**
- e. 1st host of the 1st subnet. **Ans: 201.123.101.1**
- f. DBA for the 1st subnet. **Ans: 201.123.101.31**
- g. Last host of the last subnet. **Ans: 201.123.101.254**
- h. DBA for the last subnet. **Ans: 201.123.101.255**
- i. 3rd host of the 3rd subnet. **Ans: 201.123.101.67**
- j. DBA for the 3rd subnet. **Ans: 201.123.101.95**

Problem 4: The IPv4 address and subnet mask are given as 171.24.123.146 and 255.255.252.0, respectively. Find:

- a. Number of subnets for given IP address.
- b. Number of host in each subnet for given IP address.
- c. the subnet ID for given IP address.
- d. 1st host of the same subnet for given IP address.
- e. Last host of the same subnet for given IP address.
- f. Directed broadcast address for the same subnet for given IP address.
- g. 1st host of the 1st subnet.
- h. DBA for the 1st subnet.
- i. Last host of the last subnet.
- j. DBA for the last subnet.
- k. 3rd host of the 3rd subnet.
- l. DBA for the 3rd subnet.

(Try Yourself)

Problem 4: The IPv4 address and subnet mask are given as 171.24.123.146 and 255.255.252.0, respectively. Find:

- a. Number of subnets for given IP address. **Ans: 64 subnets**
- b. Number of host in each subnet for given IP address. **Ans: 1022 hosts/subnet**
- c. the subnet ID for given IP address. **Ans: 171.24.120.0**
- d. 1st host of the same subnet for given IP address. **Ans: 171.24.120.1**
- e. Last host of the same subnet for given IP address. **Ans: 171.24.255.254**
- f. Directed broadcast address for the same subnet for given IP address. **Ans: 171.24.255.255**
- g. 1st host of the 1st subnet. **Ans: 171.24.0.1**
- h. DBA for the 1st subnet. **Ans: 171.24.3.255**
- i. Last host of the last subnet. **Ans: 171.24.255.254**
- j. DBA for the last subnet. **Ans: 171.24.255.255**
- k. 3rd host of the 3rd subnet. **Ans: 171.24.8.3**
- l. DBA for the 3rd subnet. **Ans: 171.24.11.255**

Problem 5: Let there are three IP addresses given as follows: $IP_1 = 203.117.111.144$, $IP_2 = 203.117.111.168$, $IP_3 = 203.117.111.162$ and subnet mask = 255.255.255.192. Identify which IP's belongs to same subnet.

Solution 5: Do it in class.

Problem 5: Let there are three IP addresses given as follows: $IP_1 = 203.117.111.144$, $IP_2 = 203.117.111.168$, $IP_3 = 203.117.111.162$ and subnet mask = 255.255.255.192. Identify which IP's belongs to same subnet. **Ans: All IPs belong to same subnet.**

References:

Data Communications and Networking by Behrouz A. Forouzan.



THANK YOU